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Deep Learning based medical image segmentation — Part 2 — Exploratory Data Analysis



Diya S

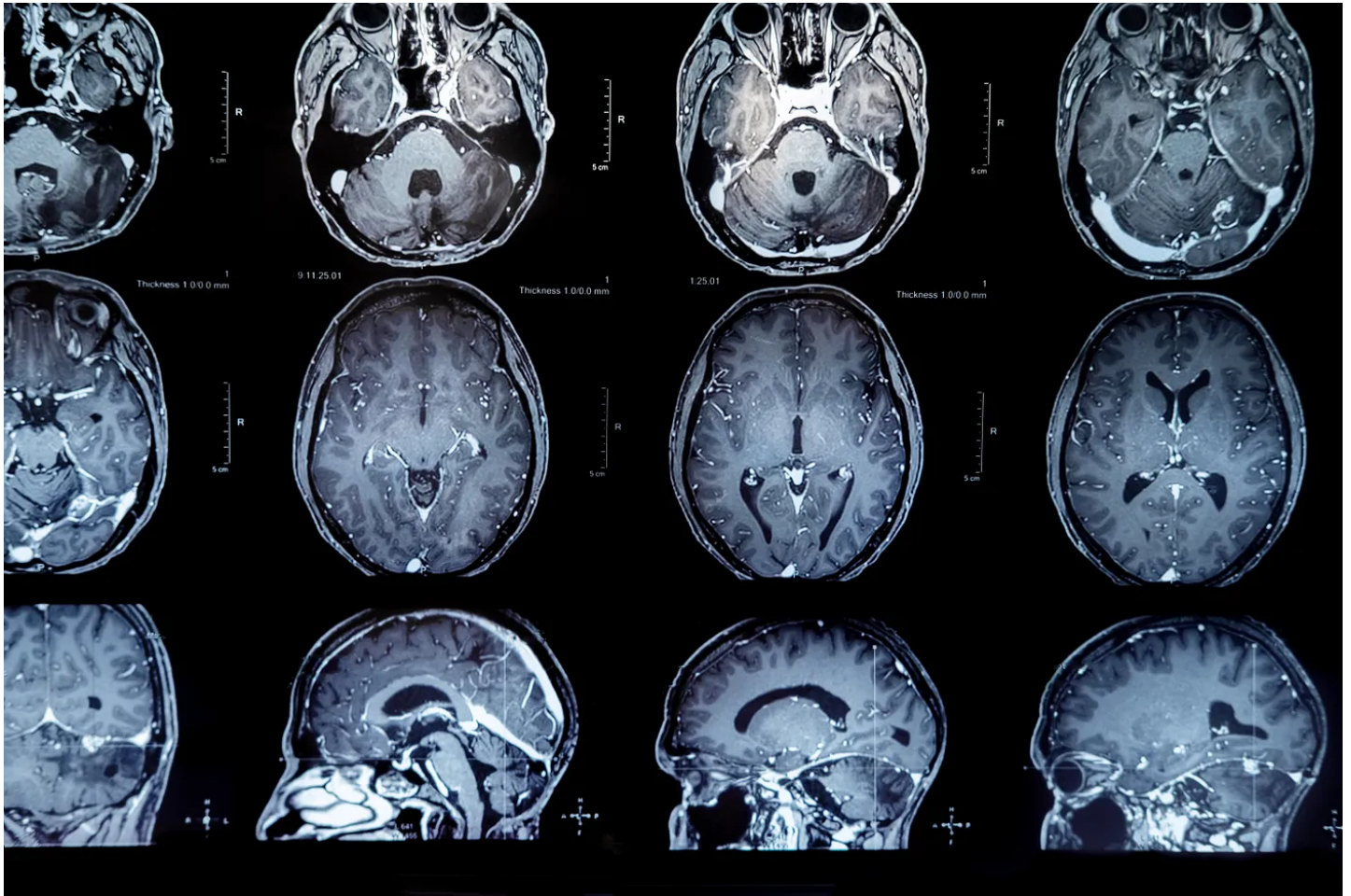
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See [Part 1](#) before diving in. We have downloaded the UPenn GBM MRI scans and the clinical data from TCIA. Let's take a look at the clinical data to get an understanding of the patient cohort.

Exploratory Data Analysis

Let's look at our patient cohort in a bit more detail. I first opened the CSV file in Excel/Google Sheets to see what it looks like.

A	B	C	D	E	F	G	H	I	J
ID	Gender	Age_at_scan_years	Survival_from_surgery_days	IDH1	MGMT	KPS	GTR_over90percent	Time_since_baseline_preop	PsP_TP_score
UPENN-GBM-00001_11	F	52.16	Not Available	Wildtype	Not Available	Not Available	Y		0 NA
UPENN-GBM-00002_11	F	61.3		291 Wildtype	Not Available	Not Available	Y		0 NA
UPENN-GBM-00003_11	M	42.82	Not Available	Wildtype	Not Available	Not Available	Y		0 NA
UPENN-GBM-00004_11	M	33.43	Not Available	NOS/NEC	Not Available	Not Available	Y		0 NA
UPENN-GBM-00005_11	M	53.33	Not Available	Wildtype	Not Available	Not Available	Y		0 NA
UPENN-GBM-00006_11	M	52.99		626 Wildtype	Not Available	Not Available	Y		0 NA
UPENN-GBM-00007_11	M	56.96	Not Available	Wildtype	Not Available	Not Available	Y		0 NA
UPENN-GBM-00008_11	F	55.06		469 Wildtype	Not Available	Not Available	Y		0 NA
UPENN-GBM-00009_11	M	69.12		561 Wildtype	Not Available	Not Available	Y		0 NA
UPENN-GBM-00010_11	M	77.54	Not Available	Wildtype	Not Available	Not Available	Y		0 NA
UPENN-GBM-00011_11	F	39.61		798 NOS/NEC	Not Available	Not Available	Y		0 NA
UPENN-GBM-00012_11	M	53.85		882 Wildtype	Not Available	Not Available	Y		0 NA
UPENN-GBM-00013_11	M	69.45		472 NOS/NEC	Not Available	Not Available	Y		0 NA
UPENN-GBM-00014_11	M	68.06		272 Wildtype	Not Available	Not Available	Y		0 NA
UPENN-GBM-00015_11	M	54.9	Not Available	Wildtype	Not Available	Not Available	Y		0 NA
UPENN-GBM-00016_11	M	70.89		509 Wildtype	Not Available	Not Available	Y		0 NA
UPENN-GBM-00017_11	M	69.23		1297 NOS/NEC	Not Available	Not Available	N		0 NA
UPENN-GBM-00018_11	F	57.21		1170 Wildtype	Not Available	Not Available	Y		0 NA
UPENN-GBM-00019_11	F	43.12	Not Available	Wildtype	Not Available	Not Available	Y		0 NA
UPENN-GBM-00020_11	M	50.86		425 Wildtype	Not Available	Not Available	Y		0 NA
UPENN-GBM-00021_11	F	49.04		510 Wildtype	Not Available	Not Available	Y		0 NA
UPENN-GBM-00022_11	F	53.88		1882 Wildtype	Unmethylated	Not Available	Not Available		0 NA
UPENN-GBM-00023_11	M	55.57	Not Available	Wildtype	Not Available	Not Available	N		0 NA
UPENN-GBM-00024_11	M	54.26	Not Available	Wildtype	Not Available	Not Available	Y		0 NA
UPENN-GBM-00025_11	F	64.32	Not Available	Wildtype	Not Available	Not Available	Y		0 NA

Clinical Data in Excel

Each patient has a unique *ID*. Their *Gender* and *Age* at the time of the MRI scan is listed. If available, the number of *Days* they survived after surgery is shown. Some patients do not have this information listed as potentially the authors were not able to contact them or collect this information before the study ended. The next two fields show the mutational status of *isocitrate dehydrogenase (IDH-1)*, and the presence of the *O6-methylguanine-DNA-methyltransferase (MGMT)* gene, both of which are strong biomarkers for GBM survival prognosis. We then have the Karnofsky Performance Status (*KPS*) score. This is a qualitative score from 0–100 that indicates the ability of the patient to perform daily activities and self-care. A KPS score of 0 means the patient is deceased. Taken together, this clinical data can help us train a survival model that can predict a patient's Overall Survival (OS).

Finally, let's write some code to analyze the clinical data. We will need some basic understanding of the clinical data so we can structure our MRI scans the right way for segmentation model training.

Login to your Google account and go to —
<https://colab.research.google.com/>.

Open a new notebook (File → New Notebook). At the top left, rename your notebook to *upenn-gbm-eda.ipynb*.

Let's mount our Google drive so the files are accessible from our Colab notebook. After running the code below, follow the pop-up prompts to select “Connect to Google Drive”, choose your account name, and click on the “Allow Access” button in the pop-up.

```
# Mount Google drive
from google.colab import drive
drive.mount('/content/drive')
```

Now let's import some basic Python analysis and visualization packages

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Read the clinical data using Pandas. Replace the file path to the CSV file with your specific directory structure if it is different from mine. Path is

case sensitive.

```
upenn_gbm_clinical = "/content/drive/MyDrive/UPENN-GBM/NIFTI-files/UPENN-GBM_cl
df = pd.read_csv(upenn_gbm_clinical)
```

You should see something like the picture below —

```
from google.colab import drive
drive.mount('/content/drive')
```

Mounted at /content/drive

```
[ ] import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

[ ] upenn_gbm_clinical = "/content/drive/MyDrive/UPENN-GBM/NIFTI-files/UPENN-GBM_clinical_info_v1.0.csv"
df = pd.read_csv(upenn_gbm_clinical)
df
```

	ID	Gender	Age_at_scan_years	Survival_from_surgery_days	IDH1	MGMT	KPS	GTR_over90percent	Time_since_baseline_preop	PsP_TP_score
0	UPENN-GBM-00001_11	F	52.16	Not Available	Wildtype	Not Available	Not Available	Y	0	NaN
1	UPENN-GBM-00002_11	F	61.30	291	Wildtype	Not Available	Not Available	Y	0	NaN
2	UPENN-GBM-00003_11	M	42.82	Not Available	Wildtype	Not Available	Not Available	Y	0	NaN
3	UPENN-GBM-00004_11	M	33.43	Not Available	NOS/NEC	Not Available	Not Available	Y	0	NaN
4	UPENN-GBM-00005_11	M	53.33	Not Available	Wildtype	Not Available	Not Available	Y	0	NaN
...	...	...	...	...	...	...	...	...	...	...
666	UPENN-GBM-00626_21	F	57.19	Not Available	Wildtype	Unmethylated	Not Available	Not Applicable	234	5.0
667	UPENN-GBM-00627_21	F	57.40	Not Available	Wildtype	Not Available	Not Available	Not Applicable	Not Available	4.0
668	UPENN-GBM-00628_21	M	55.09	Not Available	Wildtype	Not Available	Not Available	Not Applicable	Not Available	6.0
669	UPENN-GBM-00629_21	M	71.44	Not Available	Wildtype	Unmethylated	Not Available	Not Applicable	Not Available	6.0
670	UPENN-GBM-00630_21	M	65.47	Not Available	Wildtype	Methylated	Not Available	Not Applicable	Not Available	2.0

671 rows x 10 columns

We can see that the file has 671 rows and 10 columns. But from the dataset description, the patient cohort included 630 patients. So where are the additional records coming from? If you look at the image closely, you can see that certain patients had multiple scans (suffix of _11 and _21 in the ID). By looking at the data, we can further observe that the second scan, where exists, was taken later than the first (by looking at the age at scan column)

and has more information than the first. But how do we get the actual number of distinct patients? Since each patient ID ends in either _11 or _21, we can remove these final 3 characters and then get a unique count of patients, like so -

```
# Remove the last 3 characters from each patient ID, store it in a new dataframe
patients_df = df.ID.str[:-3]
patients_df
patients_df.nunique()
```

```
✓ [19] patients_df = df.ID.str[:-3]
0s

✓ [20] patients_df
0s

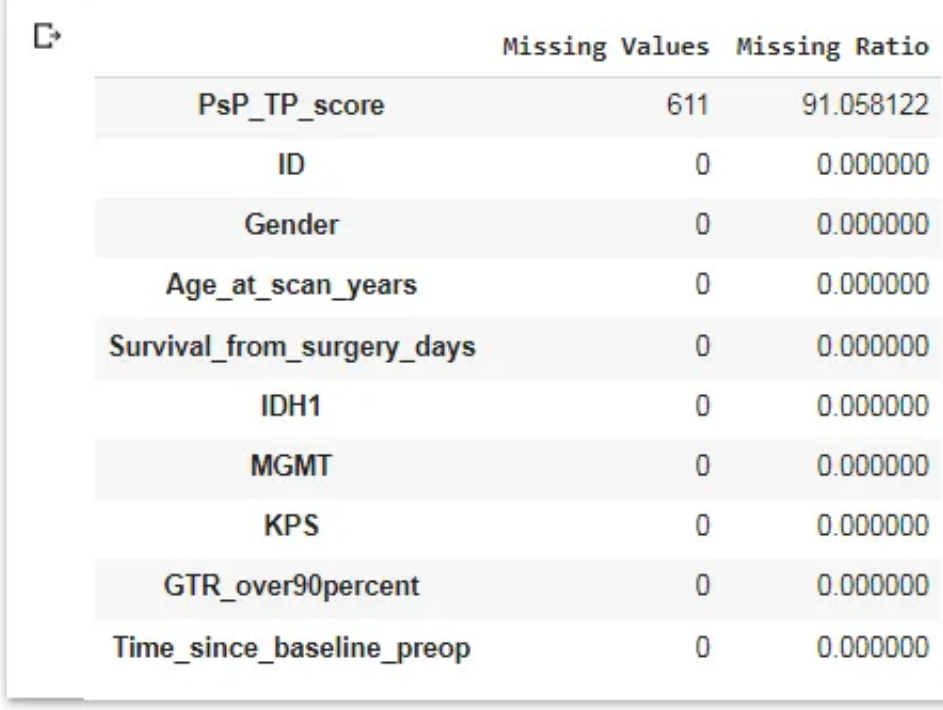
   0    UPENN-GBM-00001
   1    UPENN-GBM-00002
   2    UPENN-GBM-00003
   3    UPENN-GBM-00004
   4    UPENN-GBM-00005
   ...
  666    UPENN-GBM-00626
  667    UPENN-GBM-00627
  668    UPENN-GBM-00628
  669    UPENN-GBM-00629
  670    UPENN-GBM-00630
Name: ID, Length: 671, dtype: object

✓ [21] patients_df.nunique()
0s
630
```

We have 630 patients, which matches the number in the paper. Let's run some basic analysis on the clinical data.

```
# Check for Missing data
```

```
missing = pd.DataFrame({'Missing Values':df.isna().sum(), 'Missing Ratio':((df.  
missing.sort_values(['Missing Ratio'], ascending=False).head(10)
```



The screenshot shows a Jupyter Notebook interface with a code cell and its output. The code cell contains a pandas DataFrame named 'missing' that displays the count of missing values and the missing ratio for the top 10 variables. The output is a table with three columns: the variable name, the count of missing values, and the missing ratio.

	Missing Values	Missing Ratio
PsP_TP_score	611	91.058122
ID	0	0.000000
Gender	0	0.000000
Age_at_scan_years	0	0.000000
Survival_from_surgery_days	0	0.000000
IDH1	0	0.000000
MGMT	0	0.000000
KPS	0	0.000000
GTR_over90percent	0	0.000000
Time_since_baseline_preop	0	0.000000

We see there are no missing values in any field except the Progressive Supranuclear Palsy/Tumor Proportion (PSP TP) Score, which is similar to KPS score in assessing patient disability. We will not use this field in any further analysis.

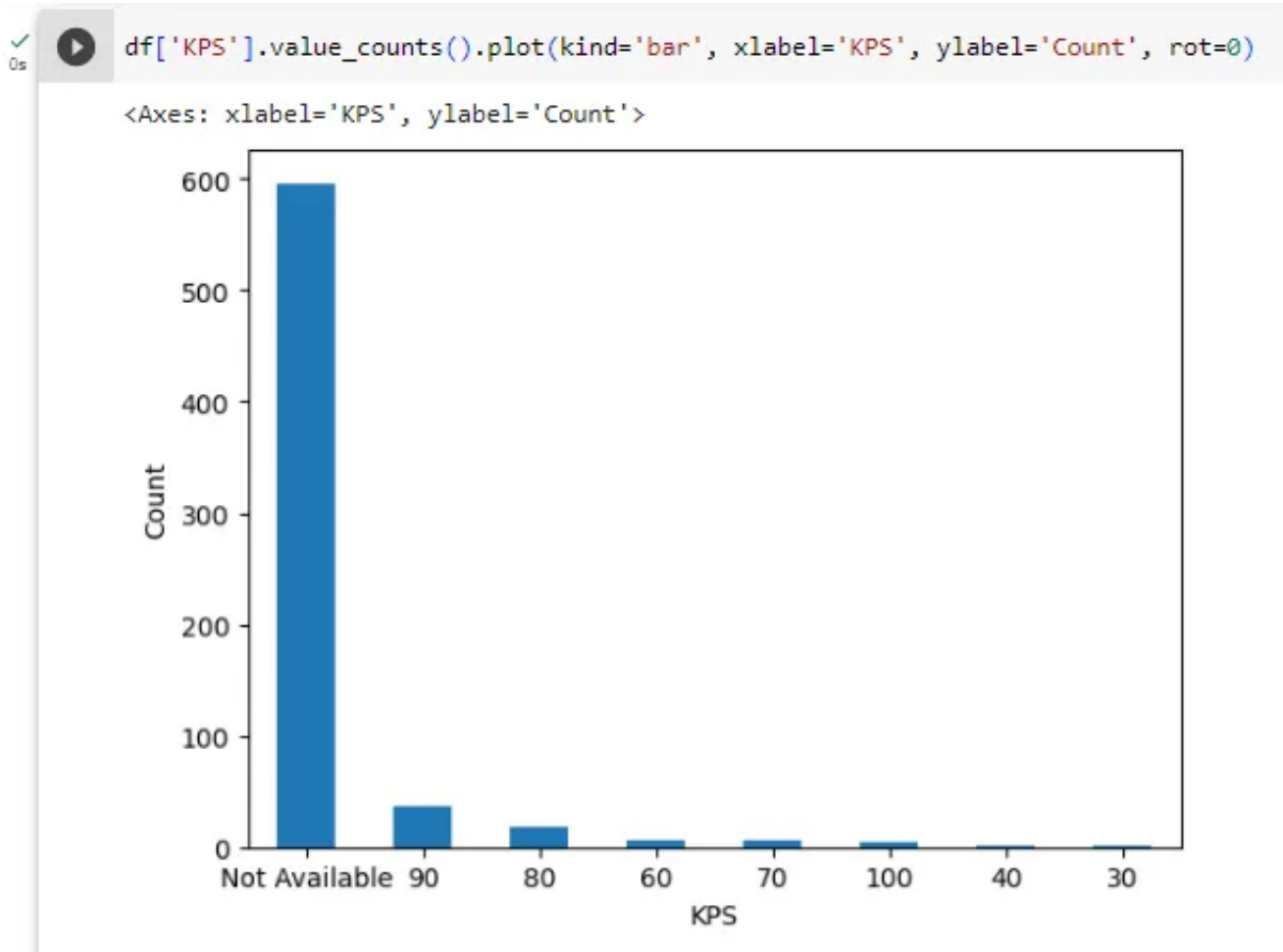
```
# Check the number of unique values in each field
```

```
df.nunique(axis=0)
```

For the categorical variables (fields those that signify a category, such as Gender), we can see what values are present in our data. For example, even though KPS is a number, each value defines a category, such as 100 = Normal, no complaints, 90 = Able to carry on normal activity with minor signs of disease and so on. We can plot a simple bar plot to see the values and counts. You can see that most of the patients do not have a KPS score (Not Available), which limits the usefulness of this field for prognosis. You can create similar plots for the other categorical variables.

```
df['KPS'].value_counts().plot(kind='bar', xlabel='KPS', ylabel='Count', rot=0)
```

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Now that we have a basic understanding of the patients in this research, we can begin the setup for training our deep learning segmentation model. Read about it in Part 3.

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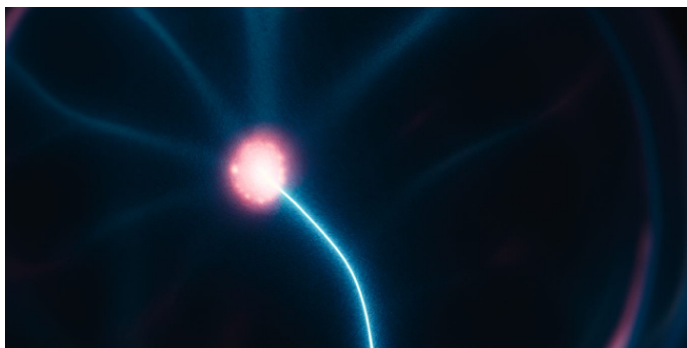
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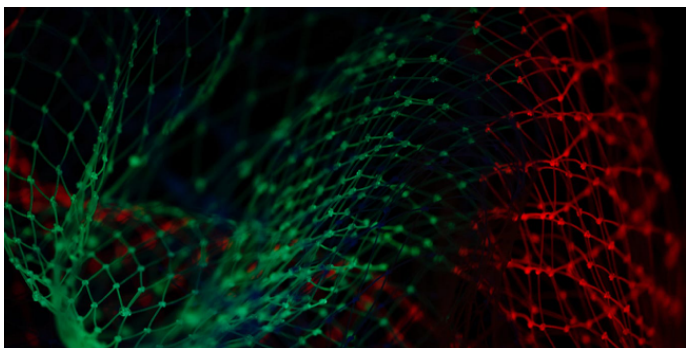
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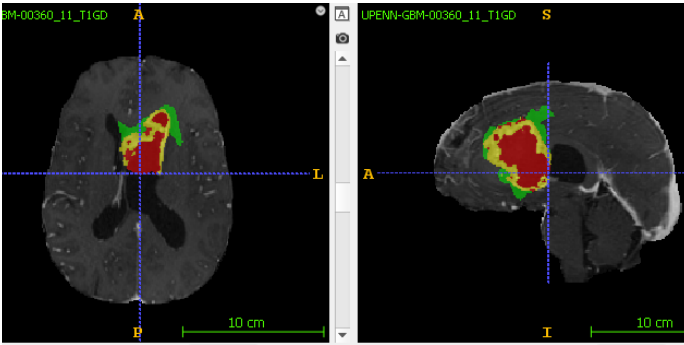
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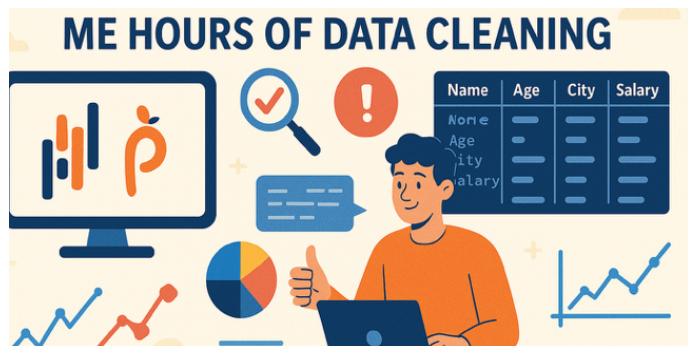
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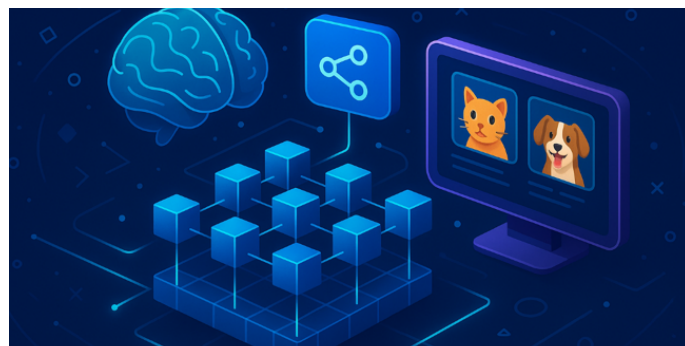


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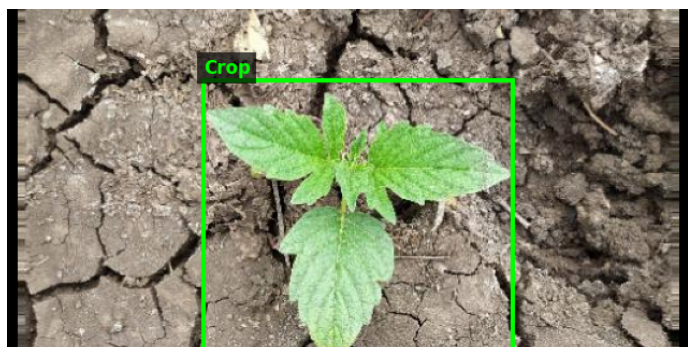


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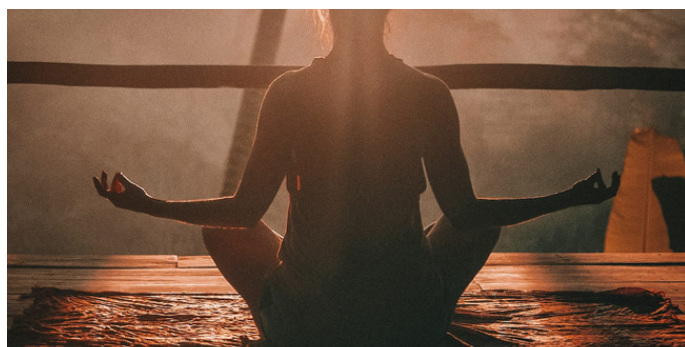
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